# Shoaib Sadiq Salehmohamed

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## Education

**Indian Institute Of Technology–Madras** — Chennai, Tamil Nadu
*Bachelor of Science (BS) in Data Science and Applications* — ***CGPA: 8.85/10*** — *May 2024 – September 2028*

**Keshav Memorial Institute of Technology** — Hyderabad, Telangana
*Bachelor of Technology in Computer Science and Engineering* — ***CGPA: 8.75/10*** — *Aug 2024 – Aug 2028*

## Technical Skills

**Languages**: Java, Python, C, SQL (PostgreSQL, MySQL), JavaScript, HTML/CSS, TypeScript
**Frameworks**: Next.js, Express.js, Vue.js, React, Node.js, Flask, FastAPI, ROS, ArduPilot
**Developer Tools**: Git, Docker, Google Cloud Platform, Azure, AWS, Linux (Ubuntu, Arch), Redis, Celery, WebSockets
**Libraries**: Pandas, NumPy, Transformers, Matplotlib, PyTorch, OpenCV

## Problem Solving & Core CS

Solved **200+ LeetCode problems** using **Java** (primary) and **C++** (secondary), covering arrays, binary search, trees, graphs, dynamic programming, and greedy algorithms; strong foundation in data structures, algorithms, object-oriented programming, operating systems, DBMS, and computer networks.

## Projects

**TattvaDrishti** | *Python, PyTorch, FastAPI, Next.js* — **SIH 2025 Winner (AICTE Cybersecurity Cell)**

- Built an end-to-end AI-driven threat intelligence platform for detecting LLM-powered misinformation and coordinated influence operations using a modular FastAPI backend.
- Fine-tuned DeBERTa-v3-Large models achieving **95% accuracy (AI vs human)** and **98% accuracy (model attribution)** on a **2.7M+ sample corpus** with **long-context inputs (12K tokens)**.
- Designed a graph-based intelligence engine to identify coordinated campaigns, narrative clusters, and influence networks across actors and regions.
- Engineered explainable risk scoring by combining **10+ linguistic, statistical, and model-based signals** to support high-stakes analyst decisions.
- Developed real-time dashboards using **Server-Sent Events (SSE)** and implemented a blockchain-based audit layer for tamper-proof logging and traceability.

**Rakshak** — **Autonomous Surveillance Rover** | *Python, Jetson Nano, LiDAR, GPS, Pixhawk*

- Built an autonomous surveillance rover integrating **LiDAR, GPS, IMU, and vision sensors** for real-time navigation, obstacle avoidance, and perimeter monitoring, resulting in a **Published Indian Patent Application** for it.
- Led development of a **hybrid control system** where a Pixhawk (ArduPilot) handled mission execution while a Jetson Nano performed perception, decision-making, and safety overrides.
- Implemented a **LiDAR-based obstacle avoidance pipeline** capable of dynamically overriding manual and autonomous commands to prevent collisions in real time.
- Engineered **real-time video streaming and telemetry pipelines** using WebSockets for remote monitoring and alert generation.
- Developed an **on-device face recognition pipeline** using OpenCV for intruder detection, enabling automated alerts and evidence capture.
- Designed a **multi-channel communication system** (Wi-Fi, RC, telemetry) ensuring robust operation in constrained and offline environments.

**Carbon Passport AI** — **Multimodal Logistics Intelligence Platform** | *FastAPI, Next.js, Graph Data, APIs, Ollama*

- Built a full-stack logistics intelligence MVP for multimodal route optimization across **truck, rail, sea, and air**, comparing carbon emissions, cost, delivery time, and operational risk.
- Designed a public-data-backed runtime logistics graph using **UN/LOCODE**, **World Port Index**, and **OurAirports** to model ports, airports, rail terminals, inland hubs, and multimodal transport edges.

- Implemented shipment-level emissions forecasting with per-leg carbon estimates, uncertainty ranges, source-status visibility, and fallback handling for unavailable external APIs.
- Added AI-assisted route explanations using local **Ollama** inference with deterministic fallback, enabling explainable route decisions without depending on cloud LLM availability.
- Built shipment passport generation with QR access and a tamper-evident audit trail for sustainability reporting, traceability, and shipment-level accountability.

**LingualLink — Real-Time Multilingual Chat Platform** | *FastAPI, Next.js, Redis, PostgreSQL, MongoDB*

- Built a real-time multilingual chat platform with automatic language detection and translation across users.
- Designed an event-driven backend using **Redis queues and Pub/Sub** to decouple message ingestion from translation workers, ensuring low-latency and scalable processing.
- Implemented a hybrid ML inference pipeline using ONNX-optimized models with GPU acceleration (FP16), achieving **0.02 sec per sentence** translation latency and **45+ sentences/sec throughput**.
- Achieved efficient language detection with **0.018 sec/sample latency** using batched inference for real-time responsiveness.
- Developed real-time communication using **WebSockets/SSE** and designed a scalable data layer with PostgreSQL (auth) and MongoDB (multi-language storage).

## Experience

**Research Intern – LLM Lens** Feb 2026 – Present

*AI Research Intern (LLM Reliability & Hallucination Detection)*

- Built hallucination detection systems across **15K+ samples (300+GB Datasets)** for LLMs using internal transformer representations and uncertainty signals.
- Designed **cross-layer and hierarchical transformer architectures** for sequence-level classification and uncertainty estimation.
- Developed **scalable data generation and evaluation pipelines** using Hugging Face and PyTorch, benchmarking with ROC-AUC, PR-AUC, F1, and calibration metrics (ECE).
- Optimized training and inference workflows on **GPU-accelerated environments (A100, H100/H200)** for large-scale experimentation.
- Analyzed layer-wise representations and covariance structures to improve model interpretability and robustness.

**Cyber Security Intern** Jun 2026 – Present

*AICTE Cyber Security Cell* *Remote / India*

- Working with the **AICTE Cyber Security Cell** to continue the development of **TattvaDrishti** into a real-world cyber intelligence application focused on threat analysis, misinformation detection, source verification, and trustworthy digital information systems.

**Member — ChatGPT Lab for Students (Inaugural India Cohort)** Aug 2025 – Sep 2025

*OpenAI* *Remote*

- Selected among **28 students nationwide** for the inaugural ChatGPT Lab India cohort.
- Explored advanced LLM workflows for academic problem-solving and research, contributing structured feedback on real-world usage patterns.

**Open Source Contributor** July 2024 – Oct 2024

*GirlScript Summer of Code (GSSoC), Hacktoberfest* *Remote*

- Ranked among **top 8% contributors** (**185 / 2613**) in GSSoC through contributions across features, bug fixes, and documentation.
- Recognized as a **Hacktoberfest Super Contributor**; contributed across repositories using **Git, pull requests, and code reviews**.

**Core Team Member – Domain Expert (Web & ML)** Sep 2025 – Present

*Recurse, KMIT Technical Club* *Hyderabad, India*

- Designed and evaluated **technical problem statements** for 24-hour hackathons involving **100+ participants**, ensuring clarity, feasibility, and appropriate difficulty.
- Mentored teams in debugging and system design, resolving **Git, tooling, and integration issues** under time-constrained environments.

- Led technical initiatives and supported development of **interactive and ML-based projects**, guiding participants in building end-to-end solutions.

**Web Development Intern** Oct 2025 – May 2026
*PR Committee, KMIT* *Hyderabad, India*

- Developed and deployed the official PR Committee website handling frontend, backend, and hosting.
- Built a **React Native + Firebase app** for event ID verification and attendance tracking.
- Designed a **scalable multi-device validation system** with real-time synchronization, preventing duplicate entries during high-traffic events.

## Publications & Intellectual Property

**RAKSHAK: Autonomous Surveillance Rover** Patent Published
*Indian Patent Application No. 202541112724 A* *Jan 2026*

- Co-inventor of a published patent on an autonomous rover integrating LiDAR, vision, sensor fusion, and hybrid control for perimeter security.

**Detecting Hallucinations via Internal Transformer Representations** arXiv Preprint
*arXiv:2604.06277* *Apr 2026*

- Proposed hallucination detection using transformer hidden states; designed and evaluated MLP, cross-layer, and hierarchical models with ROC-AUC, PR-AUC, and calibration metrics. Along with a public dataset released on Hugging Face

**BEACON: Behavioral Entropy Aggregation for Cross-Model Hallucination Detection in Large Languag**
*arXiv:2606.07528*

- Co-authored a black-box hallucination detection framework using behavioral entropy, semantic consistency, and paraphrase stability signals across LLM outputs, achieving **0.8123 AUROC** across seven benchmarks.